

EMBEDDED SYSTEMS

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PROF. DOUGLAS RENAUX, PHD

BIG IDEAS
FOR EVERY SPACE

<https://www.renesas.com/br/en/img/misc/products-solutions/synergy-ssp-title.jpg>

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EMBEDDED SYSTEMS - BASED ON CORTEX-M4 AND THE RENESAS SYNERGY PLATFORM

BY: DOUGLAS RENAUX AND ROBSON LINHARES

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AUTHORS

The authors: Douglas Renaux and Robson Linhares are faculty at UTFPR in the subjects of Embedded Real-Time Systems and Computer Architecture and Organization. UTFPR is the Brazilian Federal University of Technology.

eSysTech – Embedded Systems Technologies is a company providing engineering and training services in the area of Embedded Systems. It is a spin-off of the Laboratory of Innovation and Technology in Embedded Systems of UTFPR.

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OVERVIEW AND PREREQUISITES

This Embedded Systems course is organized into theory and practice parts. There are 12 theory sections and 9 labs. The labs solutions can be made available to instructors. All labs are conceived to be developed on the Renesas SK-S7G2 board, based on an ARM Cortex-M4F MCU.

The course assumes that the students have previous knowledge on:

- C programming for embedded systems
- Microcontrollers and assembly programming (on an architecture other than ARM)
- Digital Systems
- Digital communications and networks

1- INTRODUCTION

1. What are Embedded Systems
2. Characteristics
3. Market Segments
4. The IoT Era

WHAT ARE EMBEDDED SYSTEMS ?

An Embedded System (a.k.a. Embedded Computing System) is a computing system that is built-into (i.e. embedded) a larger device, such as an equipment, a system, or a vehicle.

Embedded Systems (ES) are usually application-specific and have real-time constraints; thus, many ES are also real-time systems. Often, ES are used in control loops: reading sensors, processing data, and generating outputs that control the device they are embedded into. Finite State Machines are commonly used to model the behavior of ES.

CHARACTERISTICS OF EMBEDDED SYSTEMS (1/4)

Typical characteristics of an Embedded System are:

1. **Microcontroller based system** consisting of a processor, non-volatile memory (Flash), volatile memory (RAM), and a large number of inputs and output interfaces as well as communication channels.
2. **Cost effective implementations** as many device architectures are cost-driven.
3. **Energy efficient solutions**, as many devices are battery powered. Current trend is to develop battery-less devices that harvest energy from the environment.

CHARACTERISTICS OF EMBEDDED SYSTEMS (2/4)

4. **Heterogeneous.** While desktop computers are based on standard platforms (Windows PC, Apple IOS, ...) there is a large variety of hardware and software for embedded systems.
5. **Variety of restrictions** to the design solutions, such as:
 - a) Physical: Size, Weight, Temperature Range, Vibration, Dust, Spills, Water;
 - b) Computational resources: processing speed, non-volatile memory, RAM, available I/O;
 - c) Response time.
6. **Interconnected.** Embedded devices and systems are ever more interconnected to each other. Trend is to increase the interconnection rate (IoT).

CHARACTERISTICS OF EMBEDDED SYSTEMS (3/4)

7. Reliability.

The ability of a system or component to function under stated conditions for a specified period of time.

8. Availability

The ability of a system or component to function at a specific moment of interval of time.

9. Maintainability

Measures how easily and how fast a system can be restored to operational status after a failure.

10. Testability

The degree to which a system or component facilitates the establishment of test criteria and the performance of tests to determine whether those criteria have been met.

CHARACTERISTICS OF EMBEDDED SYSTEMS (4/4)

11. Scalability

The ability of a system to handle increased workload by repeatedly and cost-effectively adding components to extend the system's capacity.

12. Safety

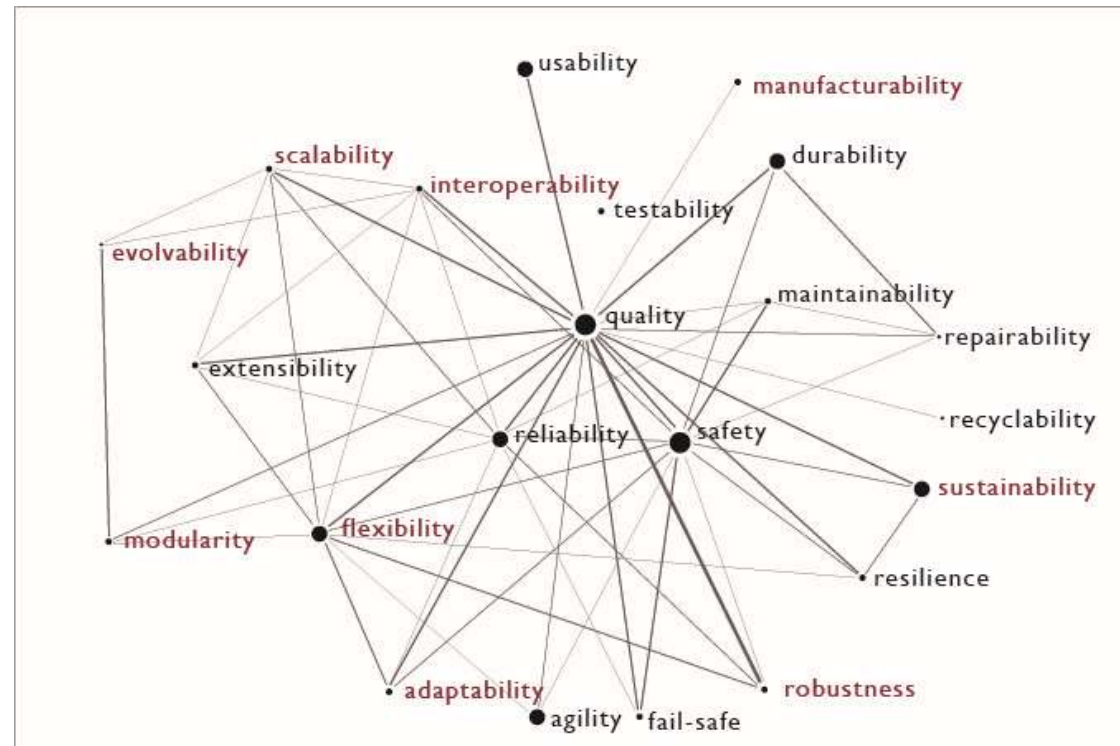
Concerns the requirement: not to harm people, the environment or other assets.

13. Security

The ability of a system to protect information and system resources with respect to integrity and confidentiality.

CHARACTERISTICS OF EMBEDDED SYSTEMS

Relationship among 22 of the most common ilities:



to read about this graph: Book Chapter about the Ilities: Chapter 4 from "[Engineering Systems: Meeting Human Needs in a Complex Technological World](http://strategic.mit.edu)" by de Weck O., Roos D. and Magee C, MIT Press, January 2012 (<http://strategic.mit.edu>)